

RMFS at a glance

Robust Multilingual Factor Score — Waiz Khan, JHU, August 2026

What it is. A report card for a model’s multilingual representations: four readings per language, each with its own validated number, scored 0 to 100 against a reference grid of 14 models. A score of 74 means better than 74 percent of the reference models.

How it is computed (about four GPU minutes per model): embed 300 parallel sentences in 41 languages at one layer, then take:

- **Meaning against language** — the share of representation structure that encodes concepts rather than language identity.
- **Sentence matching** — how reliably the model matches sentences to their English parallels.
- **Weakest aligned languages** — behaviour in the poorest aligned tail, where training damage appears first.
- **Collapse flag** — meaning structure against a reference checkpoint, for training settings.

What each reading is worth. Correlations with outside evidence, after removing training data volume, language family, script and tokenizer effects. No reading wins twice, which is why they are reported side by side rather than averaged into one score:

	carrying content across languages	downstream exams	trained variants (75 checkpoints)
Meaning against language	+0.76	+0.07	−0.10
Sentence matching	+0.76	+0.24	+0.36
Weakest languages	—	+0.09	+0.51
Collapse flag	—	—	detects exactly

Scoreboard. Qwen3-8B **90.5**, 4B **85.7**, 1.7B **73.8**; Salamandra-7B **69.0**; bloom-7b1 and Qwen3-0.6B **54.8**; OLMo-2-1B **28.6**. Scale and multilingual design orderings come out with no tuning.

Why four readings and not one number. In a study of 25 training variants (75 checkpoints), the variant trained on the word level alignment objective lost about 95 percent of its meaning structure and posted the best single score in the study; MEXA ranked it third of 25. A share cannot see a loss that shrinks both of its parts, and the same holds for any scale invariant ratio. That variant also kept its downstream ability, 0.395 against 0.390 for the untouched control, while the variants that did lose ability looked clean on every geometric measure.

What the numbers rest on. Downstream prediction sits near 0.3 after controls for every measure in this area. The 0.76 holds when whole language families are resampled (0.60 to 0.89) and rests on four models in that comparison, giving 0.12 to 0.84 when models are the unit. A panel of 69 languages carries the independent evidence of roughly two.